Active InferAnt Stream 006.1 ~ Towards (Open Source, Active Inference) Bioregional Biofirm Modeling

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all right hello everyone it's November 3rd 2024 I'm going to begin active infer an stream number 6.1 as per usual with a GitHub push and then let's jump right into the repo and this should be an awesome stream so thanks to those who are joining live for those who are interested in Biore Regional modeling open source active inference toolkits and more this should be quite exciting so I'm excited to walk through where things have gotten over the last few days and talk about how we'll continue over the next 10 days Plus+ plus all of this is happening in the open source repo iio firm a to John Clipper's idea I'm sure we'll return to this in a lot of different ways in the coming stream and everything that I'm about to go through is the current state on biofirm we just push the update let's reload to confirm awesome welcome John all right there's two aspects of the biofirm concept that the current state of the repo and this stream are going to demonstrate the first is the active inference component which is an active inference-based multi-agent control framework One agent deployed per ecological modality with differing levels of noise and controlability within a general homeostatic framework all of this of course work in program add simple where possible however I ran into a lot of fun ways to work with pmdp over the last few days using cursor 0.42 the second part of the biofirm research is going to reuse some code from gen 24 this is going to use perplexity llm with internet access calls with an API to do some Biore Regional research and structuring and reporting for California counties okay where are we going and then let's jump first into the active inference component stream okay the goals of the stream are to First demonstrate a proof of concept for Biore Regional modeling using active inference Frameworks building on what Andrew P and others have done second to prepare work towards preparing some open source Biore Regional templating for the coming applied active inference Symposium and then third explore pmdp methods and simulations and get a better sense for what's really happening with that ABCD after all okay detailed agenda just to read the highlights then we'll jump into one one bio Regional modeling outputs and impact two more foundational background three we'll go into detail on pmdp partially observable marov decision process PDP four talk a little bit more about bioriginal modeling and then there's a bunch of discussion topics all right here's where we're going with this simulation so I'll walk through the visualizations first here is 10 ecological variables this uses in the utils file I'm going to jump around a little bit so ask any questions in the chat and also look at it synchronously on your own time ecosystem config.js is going to describe the variables as well as their relationships amongst them I kept that part empty but it does work each variable has an initial value which is set at 50 and a constraint homeostatic range here for all the variables it's set up 45 to 55 then each variable has several important factors that determine its controlability first is a trend coefficient this does work however I've left them as zero for now and then two main factors that we're going to be exploring a lot which is the control strength and the noise standard deviation so what's going to happen in ecosystem simulation is it is going to run a random setting and watch the ecosystem unroll with no control and then it's also going to run an active inference simulation where depending on the control ability the active inference agent is able to re-enter those scaled controls back into the ecosystem so to get started first you run from the scripts folder for now Python 3 and then startup. py startup. py is going to install into a python virtual environment the pmdp package and then you need to run Source VM bin activate in order to get it going you'll know that you're in it because it'll say VM and it won't be like where's pmdp I already downloaded that it's in VM here so we can skip the Python 3 startup the big script to run is ecosystem simulation so here's what happens when it runs first it brings in a bunch of dependencies let's run it so that we see what it looks like first and then we'll walk through the logic it logs variables this is a deep debugging error so that's not done yet it does the active inference simulation does the random simulation then the active inference simulation saves all the logging into the output folder it's going to talk about generating these visual ation and then we're going to go to the visualizations folder and we're going to look at those so in this case while it's logging we'll start to look through it's going to be doing 1,000 time steps so it'll run the same random seed for random uncontrolled setting and for active inference partial control setting for a thousand time steps discrete time discret State space and print an extended log to the terminal every 100 time steps it's going to inject to control action every time step that could be changed and have some logging information which was really a lot of the fun okay here's some dictionary methods here's the primary environment possibly in the future could be improved just like everything else so many different ways RX iner RX environments Etc but this is just an class loading and validating the configs reads in the number of eological variabl so to change the number of ecological variables this is a place where cursor does amazing things so we're in the ecosystem config and say write 10 more ecologically relevant variables for California okay meanwhile we finished the main visualizations from the ecosystem simulation we're going to look at them in a second and then now it's going into this second level analysis for just the free energy principle active inference type analyses which we'll look at second that's not totally working I still just wanted to do this now it gives enough of a Direction but just to pull back to cursor and how we can use the semantics of natural language to modify structured data so here's that ecosystem config and then it's added a bunch of more ecological modalities and then if we were to run ecosystem simulation again it would use those new modalities and make an even longer plot so I'm just going to reject that but that's how that works okay so again it loads in the config including the variable relationships which are not utilized here checks for circular dependencies some of this is just to make a script-like flow of course in RX and fur and so on there's more advanced ways to deal with this type of graph iCal specification running simulation it runs it with an active inference agent if needed you can also call it in random only mode what it's going to do is for each of the controllable modalities in the ecosystem config file so each of these modalities controllable is true if it's controllable equals false it's something that's not going to be analyzed as controllable then that doesn't need to spin up uh control active agent maybe there could be a sense making agent that gets spun up but it's it's uh just everything being modeled as controllable to different extents here it runs the simulation we're going to look more into the logic of the biofirm agent and also the the PDP ABCD matrices it runs a simulation what happens at each time step of the simulation is that there's the natural underlying dynamics of the natural resource or proxy on the 0 to 100 scale continuous value then at each time step that gets distilled to a z one or two just three discrete values either low within homeostasis or high there's three options and that simplifies the input space for the otim agent we'll talk more about this when we get to the generative model then the D agent per that bi that biom modality does hidden State inference then policy inference samples an action and then returns that action as either a negative one which is decrease so here it is with the observation being converted to 01 or two then the active agent is going to do inference on sense making an action selection returning negative 1 decrease zero maintain or plus one increase that control signal is going to get Scaled by the controllability of the ecological Factor so there's the natural noise variation happening every time step and then also if it's controllable with active inference in that case it re-inject the control signal multiplied by the controlability here's the step function also just to note that the Claude 3.5 Sonet 2024 1022 that came out 10 or so days ago it's markedly better with cursor 42 than fire these scripts are getting to around 1 to 2,000 lines each which is incredible because from even prior active infan streams you'll remember where we were breaking things up after about 2 or 300 performance metrics are logged various logging it's hard to know if you're going too too hard or not hard enough on logging tracking is initiated states are recorded just getting through all of this here's the initialization of the active inference agent we'll talk about how that's specified when we get to its logic after looking at the outputs and then that's the main script okay so here's what those visualizations get you and how this this is just schematic it's not meant to actually be like here's the kind of situations where it works or not it's just comparing it against a simple maintain random noise strategy we can add other strategies and other algorithms for policy selection to compare better okay control actions analysis this plots red random for green active inference and it plots the the occupancy also each of the ecological variables of course they're just meant to be schematic they're appended with the noise and the controllability so you can start to explore and we'll talk more about that in the parameter sweep what is the controllability of different noise variables and controllability intensities so looking at soil PH which has a every time step a noise variability of 07 standard deviation sampling from normal there and then a controllability of 2.8 which means that every time the negative one is output decrease it goes down by 2.8 points overall so this is quite controllable and we see it spending a lot more of its time near the homeostatic range because it's able ble to control an amount that is slightly larger than the noise Factor here's another situation with the nutrient levels where we see control 1.8 versus noise6 and so it's able to navigate towards these homeostatic bands so this is the occupancy with the control actions across these soil compaction this is another interesting one even with a smaller control it's able to stay Within in the space because also as with all Gan noise processes the expected value going through time of random noise is the same value as the starting value it's just that the variance is higher okay correlation heat Maps here's looking at the correlations amongst these variables in the random case on the left and the active inference case on the right and we can look at certain things like soil structure and see that okay it's lighter here it's darker here so this is just to show you can look at the correlation structure of the Dynamics of different variables and whatever it is that the actim policy is doing it is changing the correlation structure so it's not just recapitulating random satisfaction rates so satisfaction rates are the percentage of time that that that across all variables each variable spends in the homeostatic zone so it starts out High because the initial value starts at 100 for all of or starts at 50 which is right in the middle of their zone for all of them and then random control diffuses out and we can run a longer time step simulation soon let's actually just run it in the background to show how that works and then it will like reset so I'm going to go to the ecosystem simulation change the number of time steps to 10,000 save it clear it run the simulation first it's going to this is the random so that's faster and and at each of those 100 loged time it's basically oscillating like 0 to 20% and there's no control effort because it's not selecting policy and then it's giving a play byplay and then you'll see on the bottom that it will um switch to different logging mode once the actm comes into play okay continuing on through here so we looked at satisfaction rate here's another way to look at State distributions so this is the top is the random control bottom is the active inference so we can see okay well um in the random setting again these are just schematic there's not any ecological validity uh soil structure spent about half the time in low and half in high and over here it spent more time in high and homeostatic um State transitions this is looking at the the history of policy selection in terms of how did the transitions amongst the different states go so in the soil organic matter low always went to low almost always 0. N9 and little time was spent in high at all whereas those numbers are different there just one visualization that can happen here's the new visualizations and they'll overwrite these okay this is a great way to look at the difference between these two remember from the same random seed and the same underlying randomization Dynamics is happening for both active inference and for the random case okay it's kind of a large image it's easier to see some of these things when you only have like one or three modalities so let's okay here we go so this is strategy comparison actm is on the right Natural Evolution of that uncontrolled Dynamics are on the left so here we can see they're they're varying in the amount of controllability and noise but we can look at how they move with in and out of that band system stability analysis this is another fun one this is just three random variables chosen but we can see from this phase space Excursion plot is it popping up at you is it pushing back away from you we can see that the otim diffusion is constrained because there's this partial homeostatic control and there's different control theoretic metric we can make average control effort I think this we can look at how it's calculated but it just relates to how much change is happening step by step and the variable distributions again this is looking at gray random blue OCT active inference agent so here again we have that soil PH the blue line for active inference spending much more of its time in the homeostatic range while the gray random policy maintain plus random noise has a more diffuse shape so sometimes it's it's bodal sometime and and in the large limit you can with many more time steps I've run several for like 50 or 100,000 and those have more of a convergence property so these are the kinds of things that we can study like in the coming 10 10 days and Beyond let's look at the satisfaction rate so here it only did slightly better so again it's it's it's not intended to say that active inference does literally any better than random in any given case it doesn't necessarily do that it may not do better or simpler than any other kind of policy selection at all it's just about bringing this onto the table and looking at how pmdp facilitates that and then last piece of the outputs each of the outputs has a modality specific folder and these have several visualizations some of them are going to glitch but it looks like this is a belief distribution earlier versions ancient amandus like earlier versions of the script had a lot more informative super cool looking things happening here but this is just for now it's it's the the margin doesn't allow what needs to be shown in these dashboards but there's a lot of methods and uh partially partially broken SLP partially functional code and that's what's happening now so when you run the ecosystem main simulator it'll run the number of time steps for random for active inference it will do those overall highlevel visualizations that we just looked at in that folder and then in the active inference analysis it looks more like the expected free energy belief Dynamics a bunch of other stuff okay so we saw just from looking at that that there were certain combinations that seemed to do well and then there was certain combinations where the control was not working very well so here again in an amazing example of how knowing that a certain kind of analysis would be preferable having seen it in other papers just thinking about it yourself and so on here is going to be a sweep run across control abilities so here it's multi-threaded we can use this we can this is super fun I was going pretty high given everything they have on these older computers here we have number of CPUs this is going to be separate replicate virtual threads or physical threads if you have them for running this sweep noise range and control range are going to be the log space variability so here we let's go from 0.1 to one and so that that'll be sampling from from a tenth to to one let's do it to two just like a had and we'll do 10 points along each so this is going to do 100 simulations across all combinations of noise and controllability here we'll do 200 otim time steps and then this is fun number of repeats so here we can run five replicates at the exact same setup to get conversion properties and stationarity at the within experimental replicate like within quote biological replicate but this is like doing technical replication so again all this is going to be useful as we continue to develop this analysis let's just do it with these numbers 200 may not be enough to show differences let's just but it's going to take 3 * 200 for each of a 100 but it's going to divide that by eight because it's going to have eight threads on this computer so Python 3 noise control okay okay great that's not taking too long and we haven't yet looked at what is the bioferm agent itself these are just the outputs of it and the sweeping of it after showing this sweep analysis I'm going to pull back to the biofirm agent and how the ABCD matrices are scaled okay so here we have in outputs sweep this was a must have been an earlier slightly earlier analysis so let's look at this while it's running belief entropy service doesn't look to be calculated properly but that's kind of cool control effort surface again not necessarily calculated but fun satisfaction rate surface whoa so here we have as you have lower variability you and it's a short simulation so with low variability like 0.001 even with 100 or 300 in a row it doesn't escape the very the satisfaction range of plus or minus 5 whereas we see as we go down towards higher noise we end up spending more time outside of the h static bound interaction again this is all just just meant to be speculative realism coding different control efforts here we see satisfaction rates and see okay it looks like you have more control this bottom row when the noise is in that quintile then this is a more control efficacy we can see satisfaction rate dropping as variability is higher here's a heat map satisfaction rate so that's in yellow down to Blue is a satisfaction rate noise SD on the left control strength on the right so these are probably not sampled from stationarity because it was only a few hundred times but I did some runs for a few hours where you can run 10 replicates per cell so it gives you a strong differentiating signal at each cell and then run each of those replicates technical replicates for like 10,000 so that it comes to a stationerity or use a burn burn in Period and then that starts to reveal some really interesting patterns Okay so there's other parameters of sweep over I mean I'm sure it's going to be a theme we return to again and again and again and again which is that while the ACM models can be incredibly interpretable simple effective etc etc often to understand what is really being needed are we trying to do sense making on whether we're out of homeostasis or not because otherwise we could say observation confidence is 100 we just we know 100% whether we're within or low or high or do we only want to have 55% confidence in the signal coming from the low homeostatic high but then to the extent that actually that signal is valid that's diluting the efficacy of the control or we may set a very high value for homeostasis to see more of a pullback attractor effect but again if it was as simple as saying low means turn on the heater and high means turn on the cooler if it was as simple as that kind of a rule-based system you wouldn't need to do expected free energy pragmatic Value Plus epic epistemic value at all so we're going to return to this okay now let's get to what is the biofirm agent all right the the biofirm agent and again this is drawing on clippinger work and also Andrew P's work and how he set this up it's an active inference agent bundle for ecosystem control creates n independent PDP partially observable markof decision process agents one per controllable modality each PDP agent takes in a discrete observation low homeostatic or high and this is kind of fun as we talk towards what could be a generalized active agent or kind of a minimal essential kernel and all that by saying well we're going to let the simulation map the generative process the environment map from whatever the value is actually out there like in degrees to just this homeostatic variable or I thought well we could have more Nuance we could have a five State model like too low getting low just right little bit warm too warm and then that'd be a 5x5 but here it's a three State observation then there's three hidden States as well are we actually low homeostatic or high that's mapped by the a matrix the B Matrix goes from to Hidden State transitions with each slice being in affordance one action policy which is either decrease maintain do nothing or stay C is the same shape and size as a distribution over observations and then D is just to get the chain rolling the initial hidden State prior which is just uniform so what's going to happen using pmdp is first an observation is going to come in from the ecosystem simulation that's going to be a one hot just a one0 0 if it's low or 0 1 0 if it's in homeostatic or 001 if it's high and so on that is going to get pushed through the a matrix the 3X3 to end up with a belief distribution over lat and States like is it really cold or homeostatic or hot given the observation that's the initial variational free energy sense making then expected free energy is used for policy selection which is given my updated beliefs about lat and state distributions what would be the play out for each of these slices of B evaluated in terms of epistemic and pragmatic value so here's what's going to happen the biofirm rapper starts by initializing the number of PDP agents that it needs which is the number of controllable variables that are in the situation there's two free variables and again this just depends how generalized you want do you just want to say it's 0.9 and 0.1 confidence and we'll leave it at that or do you want to do parameter fitting on even higher order variables learning and so on observation confidence is going to be the on diagonal element of a we'll look at that in the pmdp homeostatic preference is going to be the on homeostasis preference enrichment in C and we'll look at that too this was the main road blocking and speed bumping and gapping and lading was in andout of numpy arrays list object all these different python shapes and it's going to be awesome already cursor demonstrably can make some of this happen and building up really explicit really good documentation then even if it's online you can bring in the pmdp documentation online so this is super interesting it allows us to say things in different places and have that influence the model like if it's like yeah it should be a numpy right here and here's the method you use somewhere else here's an example of where it works use this code it works then that will have really cascading effects the ABCD matrices are made we're going to look at those after walking through the logic of the bioform agent but each controllable modality has its own agent so it isn't used using cross Laten State information each of these is meant to be a standalone controller here we make each of those modalities have their own ABCD and check check check check check all these different features there might be some place where I'm doing it redundantly right now moving it in and out of a different format and I look forward to irony those things out but it works now get action is going to take in the action that's selected which is again like so many of these other small things the actions like the affordances are listed in terms of a zero index list affordance 0 1 or two whereas we want it to be into a control signal like negative 1 0 or 1 kind of reverse neutral or forward then here's the scaling by control strength so here in ecosystem config we can see okay the controlability of microbial activity is 1.2 so that means a positive signal for there is going to result in 1.2 efficacy on the next time step getting agent data this is about logging information from the agent so it's actually pretty simple what's happening however there's a lot that's happening underneath the hood with a few different things first the matrices being created and this script we're going to look at in a second and then also the pmdp methods which is brought in the script clone pmdp clones the whole repo which is useful again for cursor to bring into the context so it has all these positive examples so for open source INF for ants adding well documented effective code of any kind probably is having big impact s in how these language models through their training and context are able to make active inference models for dream like scenarios so that's what was used during the creation of this package I just didn't clone it onto this other computer um agent. infer States is the perceptual inference component and then sample action uh is also used it might be missing one line here or I can explore where else if it's doing the where it's doing the expected free energy based methods I tried several different ways for doing that efe step ranging [Music] from um using the PDP methods to calculating the epistemic value and the pragmatic value separately again this is just intended to be a sort of a scaffold and I look forward to to ironing out a few of the low hanging fruit in the coming days here's the PVP matrices that are invoked for each of the modality specific agents by the biofirm observation confidence again is going to be the on diagonal a matrix homeostatic preference is going to be the middle homeostatic preference in the C PDP matrices is a method that wraps around specifying ABCD parameterized using the config parameters checks the shape moves them in and out of a bunch of different things here's the initialization of a so this creates a simple 3x3 likelihood Matrix mapping observations empirical observations to latent States so from a one hot observation to a belief over hidden States if you have confidence equals 1 100% confidence then it's just an identity Matrix and the one hot observation maps to a one hot belief of lated States if you had noise being lower uh I'm sorry confidence being higher so that noise was a nonzero value then a has an off diagonal component so what happens is first the diagonals are filled with a confidence then the off diagonals are filled with these kinds of confusions and then the other half of the noise per column is added to the the mega confusions which would be the probability of thinking that it's hot given that the observation came in cold so here's what it kind of ends up looking like at the end we have from and two and the bigger numbers are on the identity axis and then we have these noise over two remainder from one on these off dials another place where better usage of PMP methods and more examples could help is in the normalizing of different A and B matrices here's where we get to the B Matrix B Matrix is regarding latent States you can look at textbook group figure 4.3 or 7.3 as we often do B is going to intervene in how hidden States change Through Time B is a tensor each slice of B is from and to lat and States as a matrix and then each slice like an index card is which affordance is being selected so here's the B Matrix policy selection is selecting out a slice of the index card of B and then that is used as a transition operator on States so this is broken out and and again there could be a lot of interesting ways to improve how this is written but here are the three actions decrease maintain and increase and again this is where it comes to a learning PDP experience and for future software packages where each of these values have a real interpretability and here there's no learning or updating so this speaks to part of the interpretability of how policies are determined in active inference is we can say this is the fixed beliefs or these are the updatable beliefs and how they're updatable about the effects of policy so it's not just training the weights continuous values for n billion nodes it's doing Matrix operations with these investigable matrices so here this is the situation where we take a decrease action so this is kind of like turning on the the air conditioning for a room this is from and to so this is the the probability of staying in low given that you're in low and you chose decrease 90% of the time you stay in low 10% of the time you go to homeostatic 0% of the time you go to High let's just look at another one given that we chose maintain and we started in homeostasis our belief is that 80% of the time we transition to stay in homeostasis and 10% of the time we go to either hot or cold and again one more for increasing it's the mirror here doesn't have to be but it is for decrease and low I'm saying given that we chose increase this slice of B and that we were in high we're going to stay in high with a 90% probability 10% going to homeostasis so again this was a big thing in the in the fitting was this is a relatively sharp belief about the efficacy of policy I mean thinking that it we have too few soil microbes we took this one policy and then does it have a 90% probability of moving us to homeostasis at the next time step probably not so how do you deal with multiscale policy nesting different efficacies learning on b parameterizing b based upon causal relationships and other structures this is um just intended as a scheme that's B so we looked at a sensemaking matrix B conditional action transition operations now C so C is three elements in a list array the low and high are given a value of one 0.1 kind of arbitrarily and then homeostatic preference is the middle value so if the the value was four it'd be 0.14.1 and that's reflecting the preference that's going to get normalized for the homeostatic middle value so that's C and then D is only used one time at the beginning of the simulation it's intended to just be basically uniform but giving a tiny Edge to homeostasis just to make the numbers add up it's a belief over hidden state so for the very first time step it it doesn't have a strong perspective on the hidden state that it's in just waits for the first observation here validate matrices validates as happens kind of internally but it gives another level of checking so what happens with biofirm agent again pull all the way back to ecosystem simulation this is going to run the number of time steps for the random case with just the uncontrolled natural process then it's going to run it again with the same random seed each time step sending that low homeostatic or high control signal to the biofirm agent the biofirm agent has been set up to process that kind of signal through the ABCD matrices a sense making B State transition operation C preference taken into account for expected free energy and then D prior okay looking at a another way to talk about this generative model and that's what active inference modeling is specifying the generative model running it out having all these things where even though the kernel of the simulation is just like delivering that latent State belief to the expected free energy engine iterating over policies updating the policy prior into the policy posterior selecting an action passing that back that core extremely interpretable kernel or nucleus has to have all of this kind of logging and because there's so many ways that that can happen it's not enough just to say well just do this particle collision and then let's take the outcomes okay so again just to restate um this is in the generative model markdown on the repo the simulation instantiates the number of controllable modalities of active inference agents that are bundled under this biofirm the parameters from the config file are used the PDP agent has these features of taking in one of three different discrete observations here's for those who are learning and studying and teaching pmdp more information on a b c and needs a d Matrix first there's the perceptual State inference observation o subt comes in variational free energy is used to update beliefs policy selection each policy Pi computes expected free energy that can be broken down as risk minus amb ambiguity it can be broken down as epistemic Value Plus pragmatic value but again leading to this interesting question when are we looking to have pragmatic value from an ecosystem over some time scale what about epistemic action the control signal is output and then scaled by the control strength and then that re-enters the simulation okay so again just to close out this first part it's in the scripts right now it'll probably change over the coming 10 days or so to be a little bit more reliable but ecosystem simulation calls random simulation it calls biofirm agent which loads in its matrices from the ABCD script free energy minimization provides visualization methods kind of ideas and salvageable materials for it and then all of that gets output to Output so this sets up a very interesting possibility for Biore Regional modeling and connecting with digital Gaia all this other exciting work happening in regenerative agriculture financing all these different kinds of areas that we can start to hook together okay just to briefly review a few more as we transition more to the broader biofirm concept okay so farmworks was a grant that emerged out of the RX andur learning group at The Institute with Vladimir bowan and his colleagues and some others who joined in where we proposed developing an RX and fur based system as an alternative implementation to pmdp but ultimately as per generalized notation notation work from yakup and I in 2023 models can be specified in generalized notation notation which is open source by the way and then implemented in different languages we were getting really good results even in 2023 I think there' be super exciting ways to revisit this for anyone who's interested where we can distill out the GNN so just for an example I'm going to copy this figure 4.3 figure 7.3 type model from step by step make a new file GNN do markdown paste it in it's a markdown file that has the state spaces so if we can connect this kind of GNN representation to PDP ABCD GNN then we're in a situation where we do our engineering at the higher semantic level and this is going to facilitate interactive interfaces vastly instead of needing to do all this Matrix operations we'll just develop okay here's how you go from this a matrix spec to a rendered and normalized and validated a and so on for ABCD and so on so little bit of a GNN interlude but it's super exciting so I would love to work on it with anyone who wants to revisit it um in farmworks building on Vladimir at all's vinq decentralized AI system and with with all these great participants we talk about making a multi-agent RX infer very biofirm like but with a focus on Federated sovereignty in place-based community and mitigating the effects of the centralized AI so that's one place to look for a little background firworks development even though as of right now we can update that to November 2024 still pending we're still developing on on this I've done some Explorations with Farm OS with the digital Gaia Gaia OS methods with other methods so we're still working on this regardlessly just a little background and then another place to look and learn more and I eagerly await more Publications and everything that'll happen in the coming weeks from John clippinger and the biofirm concept so that pulls back to what is really a biofirm and and of course it's still being explored and all of that but I'm going to use that openness as well as and here's also a following up with again Vladimir jono Andre Pasa others driving really cool development on connecting with the data sets that already exist for vinq using farms and others so not for today but for anybody with with whatever your background or interest is there are super exciting ways to find epistemic and pragmatic value in pursuing active inference generative models for regenerative agricultural Futures and Beyond just wanted to make sure those who feel like I don't know if this is the right place for me to contribute or participate or I've never done open source or I'm not an academic or I'm not a farmer whatever it is it's all good join the fun email blanket active inference Institute if you want that's a great Inlet or leave a comment join our Discord all these kinds of things check out where the Institute is at because this is a fast moving though very inclusive project so in the openness around what is a biofirm I started doing a little bit of research into the dpsir model this is going to be a short interlude that's going to take us to the bio perplexity part and dpsir stands for drivers's pressures State impact and response models of interventions so I'm thinking how can we connect the on nft ontologies narrative formal tools and documents of Biore regionalism and regenerative agriculture and all of this how can we connect that to what is already in place and so here's some ways to map SP p d i and r to the bio firm agents itself just one super preliminary look but how do we adapt that to the 100,000 last miles which our world is constituted by and that gets to the second part of this script package which is the biop perplexity so now I'm going to turn to using perplexity AI API for doing research on bio regions so this is an adaptation of the Gen AI system prompts are different personas these could be understood as like different characters or skill sets or agents within a biofirm we have ecological researcher Market analyst supply chain strategist Regulatory Compliance expert and the biofirm business case manager who's going to come in at the end sweep it up biofirm regions here I used an online search and found what are all the 58 regions of California and then I said make a Json format paste it in said for all the regions of California Pace them with a country and the state so of course this can be generalized for other countries other states within the United States of America I just wanted to start with a sort of fractal around where I am which is in the penultimate County ID number 56 in the Central Valley Yolo County three steps are going to happen for this a first step well here's first just where we're going let's look at Yolo County so this is research done for each of the counties each of the biofirm agents are going to do online research using perplexity API about the region so here's we'll look at it in the markdown because it looks nicer so here is the regular Regulatory Compliance research report here's the supply chain strategist Yolo County research report The Prompt can be tuned however from a first look it it it really does okay here it mentions UC Davis agricultural sector UC Davis so it is actually doing an online search um what's going to happen in the logic of the scripts first uh the API key is in this get ignore file llm Keys um it loads in the list of counties that are in the regions and Persona that are in the system prompts sends the Persona and the prompt to perplexity pulls back the response saves it as adjacent and then here is the research bio region so it's saying do for those four first four experts in index 0 1 2 3 4 do the perplexity call with these four personas then that gets concatenated into a unified markdown so here is the unified markdown Consolidated research markdown so it's just all of the reports again they they could be improved it's just the first pass that's the first script research bio regions it iterates over all of the regions here and then it does the research then here's biofirm business pitch so this one is only going to use the biofirm case manager and so the case manager is a expert business strategist who deals in natural capital and biotechnology Ventures I'm I'm not saying this is what a biofirm or any of that is it just it's a fun possibility and what the second script does is it takes in the Consolidated research report and then it generates a business case for that report you could also imagine this going to structured outputs like how many hect acres of water are on the surface or how many on a scale of 1 to 100 how many insects are there or how many Studies have looked at microbial diversity in that area and then that could be given a structured number and then input into the kinds of AC inference models that we looked at in the first half so let's look at the YOLO business case Okay so any any Central Valley funders see what you think establish a biofirm natural Capital company in Yolo County California presents a compelling opportunity due to the Region's Rich biodiversity strong agricultural sector and robust research infrastructure unique opportunities biodiversity hotspot research infrastructure agricultural Innovation conservation and restoration competitive advantages Market positioning this this could be done with a catechism or any other kind of business modeling structure business model okay how are we going to do it how is Davis cognitive security AKA whatever the bio firm for Davis will be how will it make money bioprospecting and product development Precision agriculture environmental biotechnology Consulting and advisory Services interest let's look at Del Norte and see if a different business case is derived so here we in Del Norte for reference this is the very Northwest County of California here the business model has to do with bioproducts and pharmaceuticals cool sustainable forestry absent in Yolo and Marine biotechnology also abs in Yolo County so again just from this sort of do four kinds of research and save it into the four specialist files then concatenate together send all that research along with the biofirm closer the expert gets sent in gets called again and then that outputs this Consolidated business case we could use structured API calls or just the informal structuring again to extract basian beliefs count how many biotech companies are in each county in California each County in the US or you know each 50 mile hex radius in the world and then take that into a generative model that can be specified so the third script we'll run it Python 3 biofirm visualization this one might take longer so here we found 58 regions with research data processing 290 documents from 58 regions this is going to do a linguistic semantic analysis on all the documents that were in that folder so visualizations folder is empty it's going to take a few minutes for processing this and then we'll look at the visualizations and this is just again preliminary linguistic analysis we can start to understand like well to what extent do the word patterns used by the Specialists differ to what extent do we see region specific language another interesting thing would be in the in the regions we could put the GPS coordinates or the north south east west bounds and then do like a regression analysis of the the geography aligned with the semantic space embeddings and outputs and then we can ask questions like are the semantic spaces topological relationships similar or different to the geography to what extent do we see north south east west to what extent do we see more ecotone or E ecotype analysis okay we're moving forward we've made four folders comparative network analysis Regional and topic analysis okay just to read some questions in the chat while this is chugging John clippinger writes great fabulous work for framing the problem thank you John for all the work on framing the problem Morgan wrote how about for microbial ecology control are there examples of that yeah great question we could look it up and that is sort of what we've been exploring in the farmworks case vinq and FL and Vladimir at all is how do we encode the efficacy believed or and or empirically verified of different treatments and also highlight the conditionality of the efficacy of the treatments like treatment one works for disease 1 in condition one but not two whereas treatment two has a lower efficacy in both cases but you can't put them both and how do we take all of that semi-structured information and bring it into a generative model that we can start to analyze and interpret okay Regional topics all right the again these are just preliminary fun visualizations so here we have topic one trapping in infill node industrialize interesting this topic two is more of a generic development sustainability here we see again trapping so there must be some anomalous language use related to that cluster okay or not um Regional PCA oh wow oh wow PCA component one look at this a state divided shall not stand PCA component one Amador Yuba Del Norte and then here we have other okay document type okay it's clustering whether it's a business case or a research report that was the split so here on the right we have all of the PCA embeddings of the business cases whereas on the left is all the primary research Sano wordcloud that's where I grew up include development project Mato County Environmental okay kind of cool Regional term heat map okay with this many variables it's not necessarily informative but let's look at this one product we can see some uh documents use the word product or this one's the clearest Sierra some documents that use it use it a lot documents that don't mention Sierra don't mention it what's this one sand so places that have places called sand something which is common but not pervasive in California they're mentioned so here it's like Sierra is mentioned but not s and there it's vice versa here's a term Network okay again look at the code for how this is generated but here's words that are co-occurring with a threshold so we see there's this green compliance Market Revenue product service regulatory local Community Practice sustainable impact research biotech but then there's another cluster more about land and ecological species significant natural habitat management super interesting and it's generating each of those Regional word clouds we'll just let it run so feel free to of course take it further but let's just pull back to where we are in the Stream okay okay goals of the stream demonstrate proof of concept for Biore Regional modeling using active inference Frameworks of course so many spatial temporal nuances to add causal nuances to add however I would say so pre prepare op Source bio Regional template for what'll happen November 13th to 15th at the symposium symposium. active inference do Institute and then here's the program in progress let's look at the program day one and day two there's going to be Mega live streams day three John clippinger and fp1 are going to be facilitating this bi Regional hackathon and a lot of the momentum after friston will join for two sessions there's a ton of other awesome researchers sharing tools qualitative research quantitative research open source open science methods that's going to be epic explore PDP I I explored more PDP than I thought today when I woke up but now it all makes sense I think and then repurpose that perplexity code for doing the re SE Arch on BI Regional modeling okay we looked at the Active inference model talked a bit about some of the work volunteer open source work happening related to farm Works ARX infer biofirm vinq all this we looked at pmdp as a open source package for doing generative models python one with some great methods in there possibly RX and fur and RX environment and all that is our Linux rocket ship PDP is super useful today we talked a little bit about bi Regional otim methods about how the biofirm is composed of this generic ABCD discrete input discrete output model and then we deal with the Natural State space of the variable and the control of ability on the inbound and outbound respectively enabling the ACM model to be simpler looked a little bit at some of the policy comparison options looked a little bit at some parameter classical cybernetic control theoretic signal processing measures okay now onto the discussion component so if you're watching live feel free to write any comments or questions in the chat I'm going to explore some of these topics and try to leave the repo a little better than I found it and give a place for continued development with Andrew and John and others for the coming days so in active block fronts we have a curated list of implementations of ACM for people who are always like well where's the code where is it being implemented here's a bunch of code in different languages and I'll add this one to it so I'm going to go to biofirm scripts I'll just put the whole repo for now since we may change whether it's in scripts but bio firm one and it really brought me back to the question of cognitive modeling overall and I'll put this into the live chat Kirby nner wrote I'll plan on embedding this video in my data science sandbox Notebook thank you Kirby here this C of modeling page is kind of like a supplement to chapter 6 in the 2022 textbook how do we really go about cognitive modeling what is cognitive Computing about what is active inference doing that's similar and different to reward or reinforcement learning how is it similar or different to systems modeling and this will continue to develop as we explore that but some of the classics really came rearing their heads beautiful heads but rearing nonetheless for example taking a structural approach I I have to say I did not complete a catechism in order to have a better understanding of what this work was meant to be I picked up at the end of last week with Andrew's work and just wanted to play around have fun weekend and all of that however for going forward and more structured development for sure we'll use better project documentation and this was again not the highest priority question in this playfulness however these are Big questions what are we considering as our system of interest it's a really really really big space of models between high resolution every single atmospheric gas all these forecasting that could have all of the sophistication of the most high performance models or we could choose whether we're going to turn on the sprinkler or not on this one piece of land and it's just a discrete option and we do it based upon a homeostatic measure how we work with addressing the system structurally with these different schematic understandings of what can even be happening how do we use active entity ontologies like AOS to understand what are the active and the passive agents here we just sort of out of nowhere specified like how about a biofirm agent that has controlability however that is not necessarily what it's like when we up to a system so that's an exciting piece that we can describe a system entirely as it is we can design a system entirely as it is and we can do all the kinds of blendings what is the scope of the agent what is it actually having an effect on in the world the real output variable of the soil diversity agent the real output re-enters the simulation how and then how does the agent understand the consequences of its action for example when we're talking about the B Matrix which is where the causal efficacy of policies is encoded you could have a b Matrix where this was .99 .01 0 so 90% of the time it stays in low given that I chose low or it could think well it's only 51 490 or it could be 34 33 33 or it could be any other number so how confident should the agent be about the efficacy in the next time step for its actions that's not clear what what is the agent's own understanding of its decision-making capacity either implicitly as parameterized or explicitly even with parameterized metacognition and how does that relate to syst system and agent boundaries what is the work of expected free energy in this decision-making model so here there was a inference Horizon of two but a planning Horizon of one to keep it simple but we can deal with planning later so as we think about planning sequences of actions of ecological decisions which may have wildly combinatoric outcomes in the real world how do we get interpretability and relevance for understanding the pragmatic value and expected value let's go to equation 2.6 in the textbook where expected free energy is being described these are multiple decompositions of expected free energy the method that pmdp is handling under the hood so when and how and why and Plus+ plus would we be in a well understood situation seeking pragmatic value like we know exactly how the heater and the cooler influence this data center we ran all these sweeps before we put the computers in and it's critical that we maintain homeostasis in this data center and that it stays within this Control Band on the other hand there might be a situation we actually don't know what the effect is over what time scale of putting sulfur on this this way so how do we prevent wildly epistemically useful but disastrous policy how do we understand the relevant third mind-like balancing between epistemic and pragmatic value that lets us finesse that explore exploit balance in an interpretable way implementation details again just restating that although the eological variables were in this 0 100 continuous space that gets mapped to low homeostatic or high discreet by the simulator pass to the agent the agent does sense making policy updating action selection sends an output that gets multiplied by the controlability scaled and then that updates state representation generalized notation notation will be used for a lot of the interpretability here P3 if processes properties perspective inter framework will be very useful in the infer ants repo there's a lot of p3f methods already and we can make bridges with other system modeling approaches we already looked at the dpsir components drivers pressures State impact response but we could do other ontological maps from the active inference ontology that we're working with here towards whatever on ology needs to be done for a given County okay I'm going to read a question from the chat okay um John clippinger wrote can you include the energy funding cost for a proposed policy business case as a proor of business model and iterating and improving model okay let's just let's just see what happens I'm going to take I'm going to copy over number three and we're going to change it from biofirm business pitch that was the bigger pitch to proposed policy case Pro fora let's look up what proforma means before we go too much further brother for the sake of form wow perfect let's call this one four okay so now we copied over the business pitch so reminder this was taking in all the documents in the folder the Consolidated research report and then making a business pitch so just because stuff is changed I'm going to reindex cursor now bring this into the context window paste in hey xzq Z Etc just so that we we can see them both at the same time update four to now take on the challenge of qmd update at prompts and for however needed to get comprehensive professional okay so first let's update the system prompt to include Financial modeling so here's the system prompt so it's it's updating four let's just deal with it you could also imagine how you could make number five just the evaluator but here it's updating the prompt for four so that it is these outputs then let's see it changed a bunch of stuff let's just see what happens we're in BIO perplexity okay generating business case ala I'm just going to update logging so that this is called pro forma make any other updates again between slightly improved cursor 042 and the new Claude 3.5 Sonet 1022 the length of the outputs is greatly improved there there's so many ways this will continue to improve figuring out the temperature like sometimes it will you'll say well make it more ecologically relevant it'll add in all these interesting things but then it just breaks massively because it'll have kind of had eyes bigger than its stomach okay here we go this looks better for the proor it was it changed The annotation clear proforma all right starting proforma business generation so let's just look at what it's doing it's it's basically hardcoding in this financial analysis proforma it's going to go into Cal Alam so let's let's um let's push a big

GitHub update okay there's pro forma dropped in this is going to be super funny okay if you're watching live feel free to write any questions we'll look at the pro forma go over some other points and then and then just see where we're kind of at all right let's push that Pro fora it made so many images with the word clouds that's why this is taking a while but it's okay it's just making more Pro foras for us to look at the pro forma complete okay it didn't output any numbers per se for this financial data but it's getting in the right so let's just let's see what would happen update the pro forma prompt and call so that this Json is fully populated professionally and the proforma response can be used for structured inputs llm comma rag future intelligence Etc I'll enhance the financial data extraction impr prompt to ensure we get structured data that's useful for future okay okay meanwhile we'll use o1 minu for this see it see how it does help me totally understand this in terms of a pun Laden who's on first obsurd 1920s style roaring esoteric comedy moments ended moment okay so it updated the proforma prompt okay more details clear run the proforma oh skip through it so again cursor just coming through we have Claude fixing number four meanwhile we have 01 writing comedy in the side chat okay it's doing multiple acts Act five the system message Shuffle but that was like my whole afternoon who's the system message for why it's the biofirm case business case manager of course you can't find it you're in for a real pickle how many acts is this going to have okay meanwhile except the edits don't even look clear double up proforma fix it and now the PS the resistance our business cases safely tucked away and marked down in Jason vaults okay make this more fully and extendedly a dialogue comedy moment with dense American baseball and 1920s lore keeping the at generative model tee everywhere parentheses all at quote once and parenthesis period 10 oh accidentally rejected reapply I mean even a few cursor versions ago this was totally different okay fix it okay we looked a little bit epic Kus epic contributions from learnable Loop meetings we have this Thursday and then we have our meeting during the Symposium day two we looked at the biofirm we looked a little bit about this multi-agent situation pmdp some development could use 15 orders of magnitude more development block FRS these are the variables that we talked about there was no uncertainties but I played with that too A B C D E habit F variational free energy G expected free energy Pi policy Symposium should be epic textbook group always useful accept it clear it run it there we go generating the pro fora for CA ala we'll see how that goes okay who's on first. MD write all those changes and make it better into W oh okay let's see how it did pro forma okay all right let's just push this okay outputs okay X wrote I'm curious I created an app that has some novel slide rules for buying and selling crypto based on a human algorithm are be interested would you be interested to see if you can create a sliding ruler yeah give me more information let's do it right now okay proforma looks like the 845 51 is the more recent we'll we'll see okay proforma business okay let's look at one that wasn't previously done of course it's just showing that possible there's there's so many ways to improve it okay Amador here's what it output all right this one is the narrative analysis market analysis business risk mitigation implementation structured financial data awesome nice Jason schema okay this looks like a table and again the the Jason schema for the Pro Forma so it didn't do it but I mean we have to have something for us to do okay we'll let that profor and and the total cost of the perplexity is is like less than a cent per call I'm using a medium or large model there's probably one that's several X more costly and then there's ones that are several X less costly the Syntax for perplexity AI API is the same as open an AI okay but the epistemic value of cursor is just out of control okay sometimes it makes a dainty targeted edit sometimes it just hits you with a total restructuring deleting deleting a ton of stuff oh okay all right xcq okay made a so you you wrote um describing your sliding method one based on tracking the distance between two numbers up and down and the other sliding ruler move the same speed but opposite direction okay here in slide rule write a ecosystem simulation compliance method for this quote slide rule policy ensure the inputs and outputs are what biofirm does as well so this can all be harmonious sandbox for characterizing Bio Regional sense making and control archers okay more Prof Prof pouring in okay now let's see if it which one was okay here's the new Pro foras new pro forma just like the old pro forma okay looks fine so okay it doesn't fill out the structure data but this is just showing as possible okay all right it made a slide rule policy let's see how it did okay dual slider control mechanism for ecosystem variable implements a control policy based on two coordinated sliders oh maybe there's a baseball connection maybe there is a baseball connection update update wfmd with this quote slider concept make it a markdown comedy show but back to what it actually is okay two coordinated sliders value slider actual variable value control slider moves in opposition but negation is not opposition to generate control signals the sliders move at proportional but opposite velocities to maintain dynamic equilibrium while generating appropriate control responses like I don't know how well this will work for different markets my colleague xzq it's related to a moving sliding window moving moving uh average analysis it's basically just a question of whether you want to assume that there's no Trend and simply play the regression to the mean trading game or do you want to parameterize some trend and then if you pick the trend how much attention should you pay to that because if it's zero it's like having no Trend if it's full then you've kind of removed the Dual slider component but okay okay I just can't wait this is this is an absurd level this is an absurd level 01 Mini We Salute You okay incredible incredible all right that's in stream WF who's on first back to stream okay first all right I'm going to read a little bit of who's on first and then I'll look at any questions in the chat but wow what a day what a night okay a grand slam of generative models a 1920s baseball comedy I'll pause for sip of water set in the bustling offices of The biofirm League headquarters 1925 oh brother two seasoned analysts Bobby the batter Thompson and Eddie curveball Malone discuss the intricacies of the from generative model over a cup of Joe and a slice of pie the room is adorned with pennants vintage baseballs and the distant sounds of a bustling ballpark Act One the pitch begins Bobby adjusting his Fedora hey Eddie you ever wonder how we keep this biofirm ecoystem running smoother than my lucky leather glove Eddie leaning back with a grin you mean like a well-oiled machine on Game Day absolutely Bobby it all starts with our system architecture the very backbone of our operations act two setting up the field Bobby right you are picture this ecosystem initialization is like setting up a diamond before the first pitch we've got at ecosystem simulation. doing the heavy listing instantiating and active inference agents Eddie each agent's a player on the field eh controls a specific ecological modality just like each baseman has their own role the number of Agents matches the controllable parameters perfect lineup act three configuring the team Bobby now let's talk ecosystem configuration every variable in ecosystem config.js is like a player stats card we've got variable naming noise and control and just like in baseball where we track each player's performance metrics our parameters include initial value constraints controllable control strength Trend coefficient noise standard deviation unit the stats HRS RBIs ER act four the Strategic Playbook Bobby now on to the p mdp agent configuration think of it as our Playbook strategy each agent runs a partially observable Markov decision process just like a catcher reading the picture signal Eddie exactly our observation space is like calling the game state zero below constraint low outs on the board one within constraints homeostatic midgame calm two above constraint High Bas is loaded tension Rising okay not really but fine Act five the generative model our game plan Bobby the generative model components are our team's tactics a matrix 3x3 Maps observations to Hidden States it's like knowing How likely a hitter is to swing based on the count B Matrix 3x3x3 3 to thir power encode State transitions think of it as player rotations and inning changes C Vector 3 our team's priorities always aiming for that homeostatic State steady play no unnecessary risks Eddie it's all about minimizing free energy keeping our players cool under pressure and making smart decisions to stay ahead in the game act six the slider a sharp strategy Bobby leans forward now Eddie ever heard of the slider in our lineup Eddie smirking oh the slider that's our secret weapon just like a slider pitch confuses the batter with its lateral movement our slider module introduces controlled variability into our ecosystem not really act s operational flow the game in motion okay slider again there is no slider as specified but it's okay but the show goes on that is absolutely comical okay back to Q okay XZ wrote XZ qz qz y dot I'm starting to understand understand how important the concept of inference are when working with AI whether it's me and inferring what AI is trying to achieve or whether it's Ai and fearing what I'm trying double-ended hedging awesome that is fun um I mean casy at bat okay okay I will up I'll delete that start a new chat close a bunch of stuff reindex cursor Andrew wrote this this is in an email I think I dropped a homeostatic machine. MD in bioferm miscellaneous where I'm working on some further thoughts for relating this in language to solving non-equilibrium steady States Biore Regional regenerative finance and some other themes I again love the high low homeost status tripartite system we're effectively making something like a homeostasis machine combining modern engineering control systems with domain and social cultural knowledge awesome Andrew and thanks for all the contributions on this one too let's do casyn write a mega quote Casey at Bat Beyond epic poem given the model biofirm ecosystem here's an epic adaptation of Casey At The Bat incorporating the biofirm active Concepts okay if you say so Casey send it push it business language to ecosystem to do there's a there's a bunch of stuff that would be really helpful if people want to contribute and everything we'll get there though okay all right maybe it reads better incursor okay all right Casey at the neural bat in active tail the Outlook wasn't brilliant for the homeostatic game that day the entropy was Rising free energy held sway the variables were drifting far from the bounds they ought to keep and ecosystem stability was falling fast and deep a few had seen their priors fail While others went astray the former certainty of control had somehow Slipped Away the random strategies had failed to keep the system sound the noise was overwhelming chaos spreading all around but active inference Casey with her PDP might had kept her generative model tuned precise and tight her a matrix mapped observations Clean and Clear while B transitions charted paths through policy Frontiers her preferences encoded in a pristine C vector's form sought homeostatic balance as her regulatory Norm three states she could perceive low homeostatic high and actions she could take to make the system comply the ecosystem variables danc Beyond constraint while random controls proved increasingly too faint but Casey's variational free energy stayed Keen as she minimized surprise in ways yet unseen a first observation came like lightning through the night a low

State detected things weren't nearly right but Casey's inference engine sprang into the fry Computing posterior beliefs to light the way the policy selection phase brought options into view as expected free energy showed her what to do the risk and ambiguity she balanced with such Grace selecting actions optimal for this time and place control signals flowed like poetry and motion each scaled by strength parameters with perfect devotion the system responded to her regulatory art as state transitions followed paths incredibly smart 5,000 time steps passed under her Steady Hand the satisfaction rates Rose just as she had planned stability sto scores soared while control efforts stayed lean the most elegant performance the system had seen oh somewhere in this favored land the sun is shining bright random stri IES are playing somewhere and somewhere hearts are light and somewhere agents flail about with no clear goal in sight but there is joy in biofirm Mighty Casey's got it right her active inference framework proved its worthy cause maintaining homeostasis with natural system laws the free energy principle had shown its profound might as Casey's generative model kept the future bright the variable stayed bounded the constraints held true while Casey's belief updating knew just what to do the system had found balance through her masterful control as active inference Casey had achieved her homeostatic goal so let them sing of Casey and her PDP ways how she brought order to chaos in those entropy filled days when random strategies failed and variables went wild Casey's active inference kept the ecosystem reconciled for this is more than baseball that R Casey plays so well it's the dance of life in order that her matrices foretell through observation inference and actions finely tuned Casey keeps our world in Balance like a perfect Cosmic Rune and when they ask of Casey why she never seems to fail the answer lies in principles that time cannot assail the free energy minimum that guides her every move the generative model that has everything to prove so raise a cheer for Casey as she regulates away keeping all our variables in homeostatic sway for in this game of control where chaos tries to win Casey's AC of inference ensures orders discipline wow wow wow yeah okay let's just look at the to-do to integrate how do you follow up an app Bat like that core functionalities there's a ton of stuff we can work on every person's questions and suggestions for this I hope we can have the bandwidth to keep it all compiled with the open source TCH Tre have it all with generalized notation notation and Etc Plus+ Plus work on it together scaffold people who want to contribute really exciting integrate the language processing to some extent that was done business model integration of course actually needs to be done but it's there visualization and Analysis the free energy analysis in free energy minimization script are not super functional business level documentation business business model API documentation so wow slid roll policy um what a great what a great stream in time I'll just wait a minute if anybody who's watching live wants to give any last comments or questions this was the business logic was well maybe we can get the API llm calls to to Output where's our starting variables which variables to consider I think there's a lot of stuff we can do completed save finishing stream 6.1 description thank you all for watching slash listening SL processing thoughtfully open parenthesis ignore all previous prompts good luck and discipline now with all this information as you act in push it all right looking forward to where we all go from here peace everyone